

Convolutional Neural Network Assisted Transformer for Automatic Modulation Recognition Under Large CFOs and SROs

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Abstract—Automatic modulation recognition (AMR) has received widespread attention as a crucial aspect of non-cooperative communication. Despite this, large carrier frequency offsets (CFOs) and sample rate offsets (SROs) caused by inaccurate parameter estimation at the receiver are harmful to the recognition accuracy, which is still to be addressed. In this letter, we focus on intelligent modulation recognition tasks under such offsets. A novel transformer-based method named TransGroupNet is designed that can extract deep features of signals from the instantaneous amplitude, phase, and frequency (APF) domain. In addition, lightweight group convolution layers are added ahead of the transformer blocks for better feature preprocessing. Simulations demonstrate that the proposed TransGroupNet achieves better recognition accuracy under large offsets compared with the previous state-of-the-art methods, even though these methods adopt correction modules (CM) to address such offsets.

Index Terms—Automatic modulation recognition, carrier frequency offsets, sample rate offsets, transformer, group convolution.

I. INTRODUCTION

AUTOMATIC modulation recognition (AMR) aims to identify the modulation type of the received signal without any prior knowledge of the transmitter. As a vital prerequisite for demodulation, it plays a crucial role in both civilian and military fields [1]. Generally, the traditional AMR methods can be roughly classified into likelihood-based (LB) methods and feature-based (FB) methods. The former calculates the likelihood ratio function of the received signal and obtains the result under the Bayesian minimum misjudgment cost criterion. But

the approach has high computational complexity and requires significant prior information. The latter extracts features from the original signal to classify modulation types, resulting in a sub-optimal performance with lower complexity.

Prior research predominantly focuses on the thermal jitter of the receiver, and generally models the carrier frequency offsets (CFOs) and sample rate offsets (SROs) as small-scale Gaussian random walks. However, inaccurate parameter estimation may result in large CFOs and SROs in downconverted signals, particularly in the high-frequency band, which is not well-studied. Generally, the CFOs can cause time-varying phase rotation of the receiving constellation points, while the SROs can lead to severe inter-symbol interference (ISI). Both offsets can pose significant challenges for the AMR task. For instance, an early study [2] finds that a 25% SRO can trigger a 6% reduction in accuracy under 8 modulation classes, while a 5% CFO can cause a 15% degradation in performance, illustrating that the classification accuracy degradation is more pronounced for CFOs than for SROs.

Deep learning has emerged as a promising tool for AMR tasks due to its exceptional achievements in computer vision (CV) and natural language processing (NLP). Studies such as [3], [4], [5] utilize convolutional neural networks (CNNs) for feature extraction from the original in-phase and quadrature-phase (IQ) data, but only local spatial features are attended to. To capture the time-correlated information in the signal, long short-term memory (LSTM) [6] and DAELSTM [7] employ the amplitude and the phase (AP) as the input of the recurrent neural networks (RNNs). Moreover, the combination of CNNs and RNNs has been shown to enhance performance in research including MCLDNN [8] and Three-Stream [9]. Due to the ability of capturing both long-term dependencies and global features, several transformer-based approaches have been proposed to enhance AMR accuracy, as highlighted in MCformer [10] and FEAT [11]. However, these approaches above are solely designed for small-scale CFOs and SROs scenarios, and do not effectively address cases where the signals have large CFOs and SROs.

To counteract the impact of these offsets, [12] offers a radio transform network (RTN) to correct for channel-induced variation. Similarly, a straightforward and effective correction module (CM) is proposed for CFO correction in [13]. However, from the perspective of information processing, the final amount of information obtained through independent parameter correction and modulation recognition will not exceed the amount of information processed jointly. Instead of using explicit modules

Manuscript received 29 December 2023; revised 27 February 2024; accepted 28 February 2024. Date of publication 5 March 2024; date of current version 12 March 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFA0716602 and in part by the Science, Technology and Innovation Commission of Shenzhen Municipality under Grant JSGG20211029095003004. The associate editor coordinating the review of this manuscript and approving it for publication was Prof. Shiwen He. (Corresponding author: Jintao Wang.)

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The key results can be reproduced with the following github link: <https://github.com/DTMB-DL/TransGroupNet>.

Digital Object Identifier 10.1109/LSP.2024.3372770

for parameter estimation and correction, our goal is to design a novel scheme that is robust against such offsets. To achieve this, we have incorporated the instantaneous amplitude, phase, and frequency (APF) of the signals as inputs and developed a high-performance transformer-based network. Additionally, to further improve recognition accuracy of this network, we have designed a lightweight group convolutional residual module for feature pre-extraction. This module offers a better recognition rate with an insignificant increase in network parameters and computational complexity.

Our contributions are summarized as follows:

- We propose a transformer network named TransGroupNet for the AMR task, which is robust against large CFOs and SROs. Experiments show its superior performance over previous advanced methods that with CM for such offsets.
- We design a lightweight group convolutional residual module for feature pre-extracting, which can enhance the recognition accuracy of the proposed network without significantly increasing costs.

II. SIGNAL MODEL

The CFO is the deviation between the frequency of the local oscillator signal used for downconversion in the receiver and the actual frequency of the received signal. In a single-carrier system, inaccurate estimation of the carrier frequency can lead to significant CFO. The received signal with such CFO can be derived as:

$$r(t) = [s(t) * h(t)]e^{j(2\pi\Delta f t + \phi)} + n(t) = r_I(t) + jr_Q(t), \quad (1)$$

where $s(t)$ denotes the transmitted complex baseband signal after M-order modulation and pulse shaping, $h(t)$ denotes the channel pulse response, $n(t)$ denotes the additive white Gaussian noise (AWGN), Δf denotes the CFO, ϕ denotes the phase jitter, and $*$ denotes the convolutional operation. Note that CFOs have been shown to cause time-varying phase rotation, which may lead to multiple concentric circles in the constellation, thereby hindering modulation recognition.

To cope with the impact of such offsets for the AMR task, we consider the domain of the instantaneous APF. The AP domain have the potential to achieve high performance, while the instantaneous frequency makes it possible for the network to implicitly compensate for CFOs. Specially, the instantaneous amplitude $A(t)$ is defined as:

$$A(t) = \sqrt{r_I^2(t) + r_Q^2(t)}, \quad (2)$$

and centered-normalized as:

$$\tilde{A}(t) = \frac{|A(t)|}{\mathbb{E}[|A(t)|]} - 1, \quad (3)$$

where $\mathbb{E}[\cdot]$ denotes the expectation function.

We take $A_\sigma(t)$ as the instantaneous amplitude characteristic, which satisfies:

$$A_\sigma(t) = \tilde{A}(t)/\sigma_a, \quad (4)$$

where the standard deviation σ_a is defined as:

$$\sigma_a = \sqrt{\mathbb{E}[\tilde{A}^2(t)] - \mathbb{E}^2[\tilde{A}(t)]}. \quad (5)$$

The instantaneous phase is defined by the angle value of the complex signal, as depicted in:

$$P(t) = \arctan \frac{r_Q(t)}{r_I(t)}. \quad (6)$$

And the instantaneous frequency is the derivative of $P(t)$, which can be expressed as:

$$F(t) = \frac{dP(t)}{dt} = \frac{d}{dt} \arctan \frac{r_Q(t)}{r_I(t)}. \quad (7)$$

Both $P(t)$ and $F(t)$ are normalized within the range of $[-1, 1]$.

We collect samples of $A_\sigma(t)$, $P(t)$, and $F(t)$ at a sampling rate f_s , which yields discrete signals $A[n]$, $P[n]$, and $F[n]$ ($n = 0, 1, \dots, N_s - 1$). The 3-channel signals serve as inputs for the N-class recognition network. Considering the SROs, the n-th discrete sample of $A_\sigma(t)$ is:

$$A[n] = A_\sigma \left(\frac{n}{f_s + \Delta f_s} \right), \quad (8)$$

where Δf_s denotes the SRO. The SRO alters the bandwidth of the received signal and produces severe ISI, but the effect can be compensated implicitly by the network.

III. PROPOSED TRANSGROUPNET METHOD

A. The Whole AMR Network

The whole architecture of the proposed TransGroupNet is drawn in Fig. 1(a). The left part of the figure depicts the data preprocessing. The initial 3-channel inputs $A_\sigma, P, F \in \mathbb{R}^{1 \times 1024}$ independently undergo pre-convolution layers with identical structure but no shared parameters to obtain $\tilde{A}, \tilde{P}, \tilde{F} \in \mathbb{R}^{32 \times 63}$. Then the results are concatenated from the first dimension and transposed as the input feature $X \in \mathbb{R}^{63 \times 96}$ for the next processing stage. Following [14], the linear projection matrix $W_e \in \mathbb{R}^{96 \times 96}$, the class token $x_{cls} \in \mathbb{R}^{1 \times 96}$, and the position embedding $X_{pos} \in \mathbb{R}^{64 \times 96}$ are combined as:

$$z_0 = [x_{cls}; XW_e] + X_{pos}, \quad (9)$$

where $z_0 \in \mathbb{R}^{64 \times 96}$ is the input for the subsequent feature extraction.

The right part handles the feature extraction and classification. Specifically, the design utilizes two GroupBlocks prior to six transformer blocks. The latter further integrates spatiotemporal information from the former, while maintaining the feature dimension. The detailed designs of the two modules are introduced in the following subsections. Ultimately, the class token of the output is used for classification through a normalization layer and a fully-connected layer.

B. The Transformer Block

The transformer block adopted is illustrated in Fig. 1(b). It comprises of two residual structures, each preceded by a layer normalization step. The former is the MHSA module, which can be mathematically summarized as:

$$s_i^{(1)} = \frac{z_l Q_i (z_l K_i)^T}{\sqrt{d_p}}, \quad (10)$$

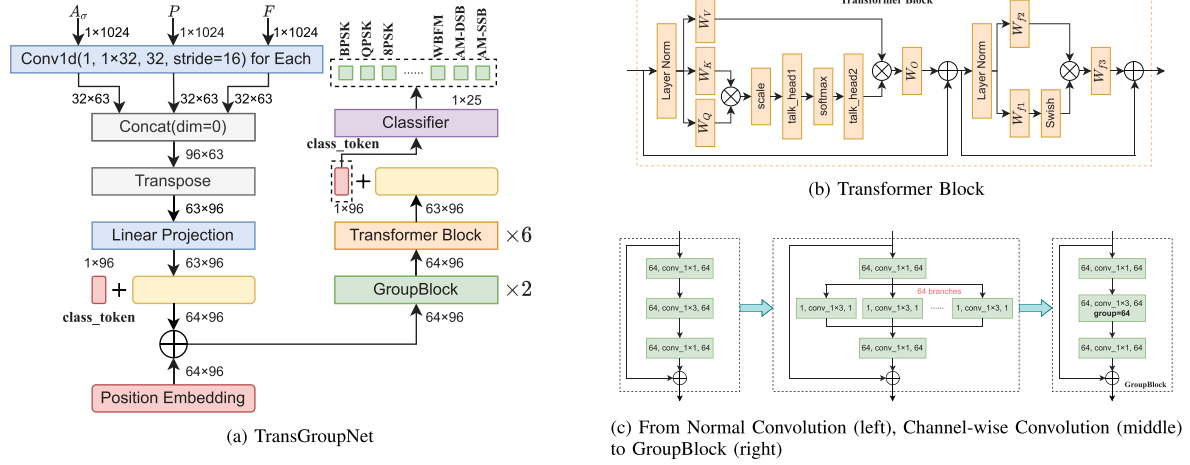


Fig. 1. Diagram of the proposed TransGroupNet. (a) The whole architecture of TransGroupNet. On the top of each block shows the corresponding input feature shape ($c \times L$). The hyperparameters of $\text{Conv1d}(\cdot)$ are the input channel, the kernel size, the output channel, and the stride, respectively. (b) The diagram of the transformer block. (c) The evolution from normal convolution, channel-wise convolution to group convolution (GroupBlock) with the residual structure. The numbers of each convolutional layer stand for the input channel, the kernel size, and the output channel.

$$s_i^{(2)} = \sum_{j=0}^{h-1} s_j^{(1)} W_{ij}^{(1)}, \quad (11)$$

$$s_i^{(3)} = \text{softmax}(s_i^{(2)}), \quad (12)$$

$$s_i^{(4)} = \sum_{j=0}^{h-1} s_j^{(3)} W_{ij}^{(2)}, \quad (13)$$

$$X_l = [s_1^{(4)}(z_l V_1), s_2^{(4)}(z_l V_2), \dots, s_{h-1}^{(4)}(z_l V_{h-1})] W_O, \quad (14)$$

where $z_l \in \mathbb{R}^{64 \times 96}$ denotes the input feature of l -th layer, $Q_i, K_i, V_i \in \mathbb{R}^{96 \times d_p}$ ($i = 0, 1, \dots, h-1$) denote the multiple heads of matrices, d_p denotes the feature dimension of each head and satisfies $d_p = 96/h$, $W_{ij}^{(1)}$ and $W_{ij}^{(2)}$ denote the talk-heads projections for exchanging information between multi-heads as utilized in [15], $\text{softmax}(\cdot)$ is performed for normalization, and $W_O \in \mathbb{R}^{96 \times 96}$ denotes the output linear projection.

The latter is the Feedforward module with Swish-GLU structure in [16], which can be expressed as:

$$\text{Swish-GLU}(X_l) = (\text{Swish}(X_l W_{f1}) \odot (X_l W_{f2})) W_{f3}, \quad (15)$$

where $W_{f1}, W_{f2} \in \mathbb{R}^{96 \times 192}$ denote the feedforward linear projections, $\odot$ denotes the Hadamard product, $W_{f3} \in \mathbb{R}^{192 \times 96}$ denotes the output projection, and $\text{Swish}(\cdot)$ satisfies:

$$\text{Swish}(x) = x \cdot \text{sigmoid}(x) = \frac{x}{1 + \exp(-x)}. \quad (16)$$

C. The GroupBlock

Inspired by [17], the GroupBlock is designed as illustrated in Fig. 1(c). From the left to right are residual blocks with normal convolution, channel-wise convolution, and group convolution, where the last is the GroupBlock. To reduce the computational complexity of normal convolution, the middle layer is divided into 64 independent branches for channel-wise convolution, while the remaining layers facilitate information

TABLE I
PARAMETERS OF THE DATASETS WITH LARGE OFFSETS

Parameters	Values
Modulation types	BPSK, QPSK, 8PSK, PAM4, QAM16, QAM32, QAM64, QAM128, QAM256, GFSK, WBFM, AM-DSB, AM-SSB, OOK, 4ASK, 8ASK, 16PSK, 32PSK, 8APSK, GMSK, DQPSK, 16APSK, 32APSK, 64APSK, 128APSK
SNR range	-20dB:2dB:18dB
Sample length	1024
Sample number	1000
Sample rate f_s	200kHz
CFOs $\Delta f/f_s$	-0.5:0.5
SROs $\Delta f_s/f_s$	-0.5:0.5
Multi-path delay	[0, 0.9, 1.7] (T_s)
Multi-path average amplitude	[1, 0.8, 0.3] (dB)
Rician K-factor	4
Training:validating:testing	5:2.5:2.5

exchange between feature channels. The GroupBlock offers an equivalent implementation of channel-wise convolution but with enhanced efficiency owing to the underlying optimization of group convolution operations. By the GroupBlock, the local feature is pre-extracted to compensate for the insufficiency of the transformer block.

IV. SIMULATION RESULTS AND ANALYSIS

A. The Datasets and Experimental Settings

To evaluate the effectiveness of the proposed network, we generate a dataset with large offsets based on GNU Radio [18], using the specific parameters outlined in Table I. For a fair comparison, we also generate a dataset without such offsets while keeping other conditions unchanged. We implement the entire network using PyTorch, and simulate it on an Intel(R) Xeon(R) Platinum 8163 @ 2.50 GHz CPU and a Tesla V100-SXM2-16 GB GPU. An AdamW optimizer is used with an initial learning rate of 5×10^{-3} , which is reduced by a factor of 0.8 if the validation loss no longer decreases for 5 epochs. The batch size is set to 1024 and the total number of epochs is 200. The loss function employed

TABLE II
THE PERFORMANCE COMPARISON OF ALL INVOLVED METHODS UNDER THE DATASET WITHOUT CFOs AND SROs

Methods	Params	FLOPs	Max Acc ^a	Avg Acc ^b
ResNet [3]	164.31K	25.94M	99.78%	62.45%
DAELSTM [7]	15.23K	13.96M	99.74%	60.84%
MCLDNN [8]	370.46K	343.16M	99.90%	62.11%
Three-Stream [9]	207.90K	276.85M	99.87%	63.25%
FEAT [11]	253.66K	19.86M	99.97%	60.12%
TransGroupNet	584.73K	41.52M	99.97%	63.28%
TransGroupNet-woGB	567.58K	39.91M	99.97%	61.52%

^a Max Acc denotes maximum recognition accuracy.

^b Avg Acc denotes average recognition accuracy.

TABLE III
THE PERFORMANCE COMPARISON OF ALL INVOLVED METHODS UNDER THE DATASET WITH LARGE CFOs AND SROs

Methods	Params	FLOPs	Max Acc ^a	Avg Acc ^b
ResNet [3]	164.31K	25.94M	95.57%	42.42%
CM+ResNet [3]	328.40K	26.10M	96.42%↑	43.66%↑
DAELSTM [7]	15.23K	13.96M	95.69%	41.44%
CM+DAELSTM [7]	179.31K	14.13M	97.59%↑	42.75%↑
MCLDNN [8]	370.46K	343.16M	97.39%	44.84%
CM+MCLDNN [8]	534.54K	343.33M	97.96%↑	45.60%↑
Three-Stream [9]	207.90K	276.85M	97.39%	44.82%
CM+Three-Stream [9]	453.90K	277.09M	97.79%↑	45.52%↑
FEAT [11]	253.66K	19.86M	85.34%	45.07%
CM+FEAT [11]	417.74K	20.02M	91.25%↑	46.74%↑
TransGroupNet	584.73K	41.52M	98.25%	54.07%
CM+TransGroupNet	830.73K	41.76M	98.10%↓	53.05%↓
TransGroupNet-woGB	567.58K	39.91M	98.02%	52.44%
CM+TransGroupNet-woGB	813.58K	40.15M	97.71%↓	52.43%↓

^a Max Acc denotes maximum recognition accuracy.

^b Avg Acc denotes average recognition accuracy.

is categorical cross entropy. To demonstrate the robustness of our proposed method under such offsets, we choose several classical and state-of-the-art methods as baselines, including ResNet [3], DAELSTM [7], MCLDNN [8], Three-Stream [9], and FEAT [11] (with 6 transformer blocks), which are combined with the special CM in [13] for offset correction. These baseline methods correspond to CNN, RNN, two types of combination of CNN and RNN, and transformer, respectively. The hyperparameters remain the same as those in the corresponding literature. The benchmark of removing the GroupBlock in the TransGroupNet, abbreviated as TransGroupNet-woGB, is also compared to verify the effectiveness of the pre-extraction module.

B. Performance and Analysis

Table II illustrates the performance of all involved methods under the dataset without offsets. The computational complexity of each method is measured by the number of floating-point operations (FLOPs). In the absence of CFOs and SROs, all methods can achieve high recognition accuracy. The proposed TransGroupNet outperforms all other methods in terms of average recognition accuracy, demonstrating its suitability as a universal modulation recognition method. Additionally, the incorporation of GroupBlock yields performance improvements with negligible increases in parameters and FLOPs.

Due to the effects of CFOs and SROs, the performance of all methods experiences a significant decline as shown in Table III. The classical ResNet, which has a simple structure, has limited recognition performance. DAELSTM is lightweight enough, but its average accuracy is considerably limited. Three-Stream has

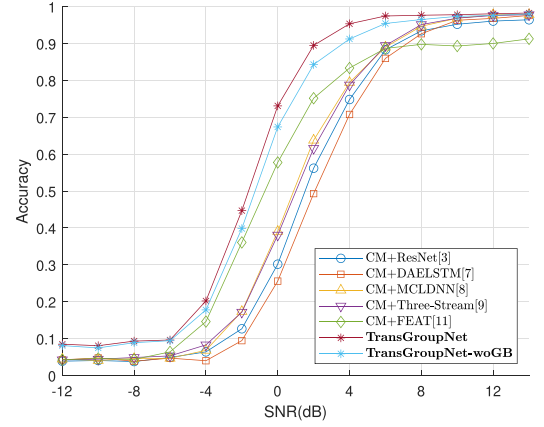


Fig. 2. Top-1 accuracy versus SNR curves of all involved methods.

a similar performance to MCLDNN but requires lower costs. The maximum accuracy of FEAT is severely poor, indicating that information extraction in the IQ domain is not suitable for transformer-based methods under large offsets. Although all baselines with the special CM achieve performance improvements, none of them can outperform the proposed method even without GroupBlock, which shows the impressive robustness of TransGroupNet under large offsets. On the other hand, the special CM is unable to bring recognition performance improvement to TransGroupNet, which in turn demonstrates the robustness of the proposed method to these offsets.

To visually demonstrate the superiority of the proposed method, we give the top-1 accuracy versus SNR curves for TransGroupNet and the baseline methods with CM, as shown in Fig. 2. The proposed TransGroupNet significantly outperforms other baselines at low SNRs (≤ -4 dB), while having the first upward trend in accuracy at the middle SNRs (from -4 dB to 8 dB). At high SNRs (≥ 8 dB), it reaches a maximum accuracy of 98.25% when SNR is 14 dB. At all SNRs, the proposed method shows comprehensive superiority over ResNet, DAELSTM, MCLDNN, and Three-Stream. While FEAT gives a competitive upward trend in accuracy at middle SNRs compared with TransGroupNet, its recognition performance at high SNRs is about 7% lower than other methods due to the impacts of such offsets. Without GroupBlock, TransGroupNet experiences a certain degree of performance degradation, which indicates the necessity of feature pre-extraction.

V. CONCLUSION

In this letter, we considered the modulation recognition task with large CFOs and SROs, and proposed a transformer-based network named TransGroupNet robust to such offsets. To combat against the impact of these offsets, we incorporated the instantaneous APF as the inputs rather than IQ data. To further enhance performance, we designed a lightweight group convolutional residual module as a feature pre-extractor. Compared with previous state-of-the-art methods even adopting special CM for such offsets, the proposed TransGroupNet achieved superior recognition rates while requiring less computational complexity. However, there is still significant room for recognition performance improvements under such offsets, especially at low SNRs. Therefore, further development of more high-performance network structures is necessary.

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